TONGFEI (SOPHIE) ZHANG

Montreal, QC +1(438) 929-1016 tongfei.zhang@mail.mcgill.ca

EDUCATION

Bachelor of Electrical Engineering—McGill University SEP 2020 - MAY 2025 (expected)

- CGPA 3.2/4.00
- Relevant Courses: Power Engineering and System Analysis, Communication System&Networks, Electronics & Microelectronics, Electric Circuits, Field & Magnetic Field, Computer Organization, Numerical Method, Microprocessor

ENGINEERING PROJECTS

Treasure Hunter (C) Dec 2024

- Designed a 2D array-based grid puzzle and integrated UART feedback for directional hints and accelerometer -based movement controls from Microprocess STM32
- Implemented audio from the speaker for clues and traps, threading tasks for audio, movement, and feedback.
- Enhanced gameplay experience with trap mechanisms, a reset button, and a step-based feedback system from OS.

DAC, Timers, Interrupts, DMA, and Analog Interfacing (C&Assembly) Oct 2024

- Configured STM32 DAC to generate wave signals for audio output, and make them observable on a speaker.
- Utilized CubeMX and HAL libraries to initialize DAC output and synchronize with timer interrupts
- Implemented DMA to optimize signal processing, and reduced CPU load for efficient multi-tone generation.
- Enabled user interaction with push-button interrupt to provide adaptive audio feedback in embedded systems

Design front-end for conscientious consumers of McGill(JavaScript) JAN 2024-NOW

- Enhanced Stocate's user experience by aligning consumer journeys with optimized UX features.
- Key tasks include mobile optimization, implementing a design system for UI consistency, conducting usability tests, and automating front-end testing. By using PostgreSQL, C# API, and Visual Studio on Google Cloud Platform, managed via GitHub Actions CI/CD pipelines.

ParkinSync-McGill BioMechanical Club (Bio-Electric Stream Captain) NOV 2023-MAY 2024

- Defined key tremor parameters and selected appropriate IMU sensors to capture essential data, isolating involuntary ($f > 3Hz$) from voluntary motion ($f < 2Hz$) while minimizing noise.
- Designed a computational algorithm to simulate system response, establishing and validating metrics
- Sourced materials and prepared construction tools with a structured timeline for prototype development.
- Integrate all electrical connections to complete the bio-electrical system configuration

IoT Smart GreenHouse MAY 2023-FEB 2024

- Created wiring schematics on LTSpice and prototyped the connection of switches and sensors on breadboard.
- Designed waterproof enclosure for Raspberry Pi and hardware components on Inventor CAD.
- Flashed a Raspberry Pi Pico to ensure network communication with the server over MQTT.

Cube Sorting and Delivery(Python) JAN - MAR 2022

- Programmed the robot using Raspberry Pi to deliver cubes based on customer instructions.
- Developed and assembled a robot with LEGO components and sensors to sort and deliver color-coded cubes
- Delivered the corresponding color cubes to the designated area based on customer pre-instructions.

EXPERIENCES

SE&I GCN Intern —*Ciena, Montreal* JAN-APR 2024

- Designed proper optical testing plan and tested required switches and optical components in Ciena Labs
- Proficiently using APIs to connect to different servers and shelves to retrieve measurement data.
- Ensure that the bonus and commission system works properly with updated data and retests to confirm.
- Developed documents for product introduction, including testing data and instruments, and updated customers' back-office network management tools to align with the new products' performance

Lambda Gain Engineering Intern—*Fonex Data Systems Inc, Montreal* MAY -DEC 2023

- Assisted in the research, analysis, and testing of optical products including transceivers and passive components
- Created Engineering Resources including EEPROMs, schematics, testing guides, etc
- Conducted acceptance testing to verify the conformance of optical networking products to industry standards.

Operational Assistant— *Easy Edu McGill, Hybrid* JUL 2020 - SEP 2021

- Planned hosted and promoted the company's activities through social media and offline.
- Communicated with stakeholders to complete the operation process.
- Responded promptly to inquiries from students and parents, organizing information systematically.

SKILLS

- Programming: JAVA, JavaScript, C, Python, Assembly, VHDL, Front-end Design
- Software & Hardware: MATLAB & Simulink, LTSpice, IntelliJ, Thonny, IDLE, FPGA, Video-Editing software, CAD, Viamaker, Poster-Producing software, Github, Visual Studio, MS Office Suite-Word, Excel & PowerPoint
- Languages: English (Fluent), Chinese (Native), French (Intermediate)